Conversation Intelligence Copilot

Detailed engineering report · generated 22 October 2026 from the repository's data, manifests and evaluation results.

Executive summary

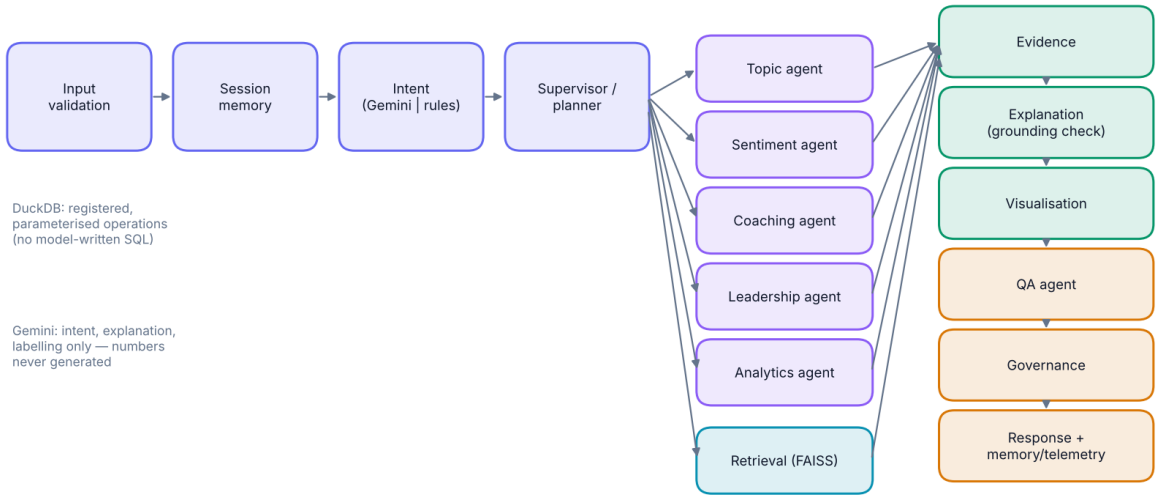
The copilot answers contact-centre questions through a supervised LangGraph workflow over deterministic DuckDB analytics, FAISS retrieval and session memory, with Gemini for language and classification when available. All four assignment questions and the three bonus extensions work end to end in the API and the UI; every answer carries evidence linked to source rows, uncertainty notes and a QA result.

Measure	Result
Q1 vs independent oracle	exact match
End-to-end QA pass rate	100% (51 requests, 11 follow-ups)
Unsupported refused / supported answered	6/6 / 45/45
Intent (rules) tuning / held-out	100% / 50%
Retrieval topic P@5 (LSA)	36.7% (random 12.5%)
Latency p50 / p95 (deterministic)	159 / 602 ms
Gemini	not reachable: GEMINI_API_KEY is not set

1. Requirements

Requirement IDs (FR, NFR, GOV, EVAL) are mapped to modules and tests in docs/requirements/REQUIREMENTS_MATRIX.md; per-feature status in docs/IMPLEMENTATION_MATRIX.md. Scope is separated into assignment-required (four questions, coordinated analysis, explainable results, documentation), bonus (retrieval, specialised agent, memory) and enterprise enhancements (nine-page UI, harness, prompt library, telemetry, governance, human review, dataset modes).

2. Architecture



Layer	Responsibility
Data	schema discovery, filler removal, PII masking, quality report, atomic DuckDB build
Labelling	versioned taxonomy, deterministic labeller, resumable Gemini labelling with cache
Analytics	13 registered operations; Wilson CI, chi-square, z-tests; sensitivity analysis
LLM	structured output, repair, rate limit, retries + Retry-After, circuit breaker, cache, token/cost accounting
Retrieval	chunking, embedder adapter, FAISS + metadata filters, evidence assembly
Agents	LangGraph graph, supervisor, specialists, explanation, QA, governance, tracing, checkpointing

Layer	Responsibility
Memory / telemetry	SQLite sessions with deterministic references; requests, node events, LLM calls, audit
UI	React chat workspace, dashboards, explorer, harness, prompts, metrics, governance

3. Data profiling and transformation

Measure	Value
Messages	41,965 (customer 21,639, agent 20,326)
Conversations	3,000 · 10–18 messages
Agents	2,987 (max 2 conversations each)
Time range	2025-08-01 10:00:00 – 2025-12-02 10:49:00
Unique cleaned texts	426

Check	Count	Severity
duplicate_messages	0 / 41965	pass
duplicate_turn_index	0 / 41965	pass
duplicate_conversations	0 / 3000	pass
missing_timestamp	0 / 41965	pass
missing_turn_index	0 / 41965	pass
unknown_speaker	0 / 41965	pass
empty_text	0 / 41965	pass
missing_agent	0 / 41965	pass
turn_index_gaps	0 / 3000	pass
timestamps_not_in_turn_order	3000 / 3000	warning
multiple_agents_per_conversation	0 / 3000	pass
not_customer_opened	0 / 3000	pass
very_short_conversations	0 / 3000	pass
length_outliers	0 / 3000	pass
synthetic_filler_tokens	41965 / 41965	info
unsupported_characters	0 / 41965	pass
pii_detected	0 / 41965	pass

Transformation: canonical schema → clean_text (filler removed only when the tail is pure lowercase filler) → masked_text → stable IDs and source line references → labels → conversation aggregates. Analytical decisions are documented in ASSUMPTIONS.md (dataset interpretation, labels, windows, scoring).

4. Labelling methodology

100% of original messages match a template (template coverage {'template': 1.0}). Topic/product come from the opening template; sentiment scores are an ordinal mapping of the 18 recurring phrases; conversation sentiment uses mean thresholds ±0.25 with a 'mixed' class. Unknown phrasing receives a provisional lexicon label (confidence 0.4). Lexicon fallback vs template labels on 21,639 customer messages: accuracy 78.1%, κ 0.6692. Human review: no_reviews. Gemini labels: not run (model unavailable).

5. Gemini integration

google-genai client with response_schema (Pydantic, closed enums), temperature ≤ 0.2, thinking budget 0; one repair attempt with the validation error; sliding-window RPM limiter and concurrency cap; retries only for 408/429/5xx/timeouts with full-jitter backoff or the server's retryDelay; permanent auth errors open the circuit; response cache keyed by model, prompt id+version, system, user, schema and config; per-call usage (streamed usage treated as cumulative), latency, retries, cache hits, cost estimate. Verified against a scripted SDK; the live API returned 401 API_KEY_SERVICE_BLOCKED in the development environment.

6. Agent harness and orchestration

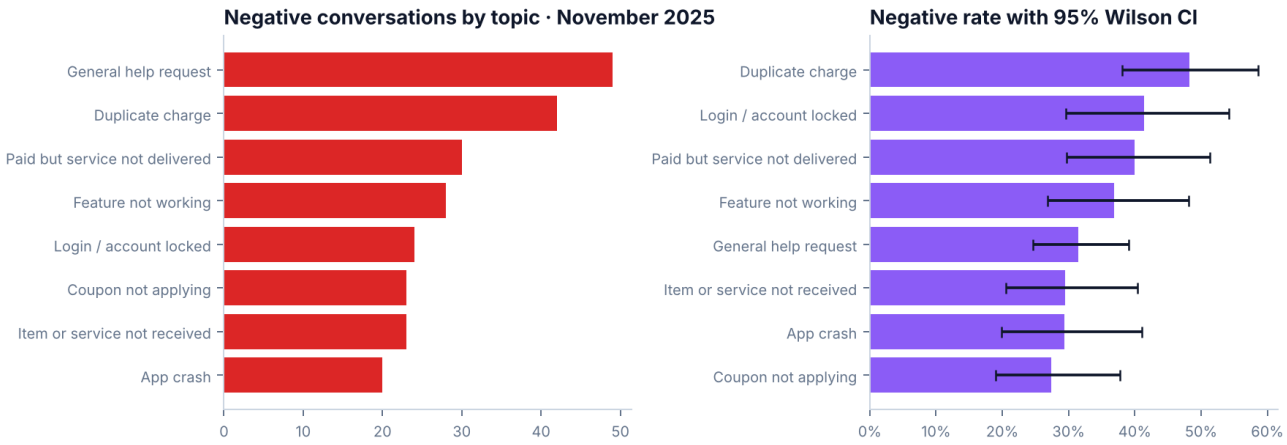
Conditional routing to one specialist; retrieval in parallel (same superstep) or after analysis; ≤ 4 operations per request; 90 s budget; node failures recorded without crashing non-critical paths; client disconnect cancels at the next node boundary; LangGraph SQLite checkpoint per request.



Agent Harness: real node statuses, durations, skipped nodes and the node inspector.

7. Analytical workflows

Q1 — negative topics

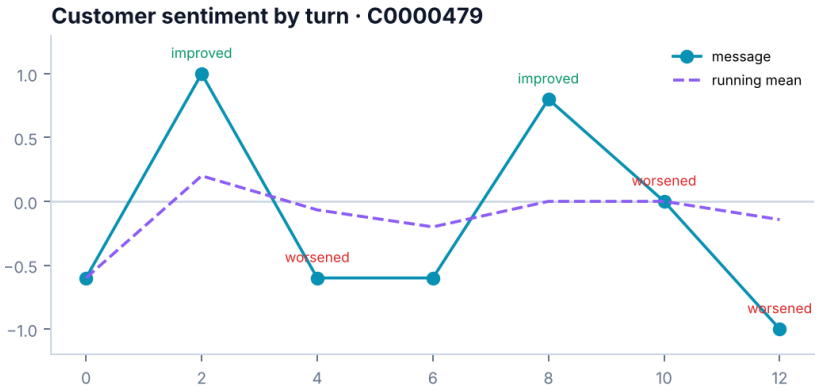


The negative rate does not differ significantly across topics (chi-square p = 0.0566). Rankings mostly reflect how many conversations each group has, so treat small gaps between ranks as noise. Sentiment and topic labels are derived from message templates by a deterministic rule set (see EVALUATION.md for agreement checks).

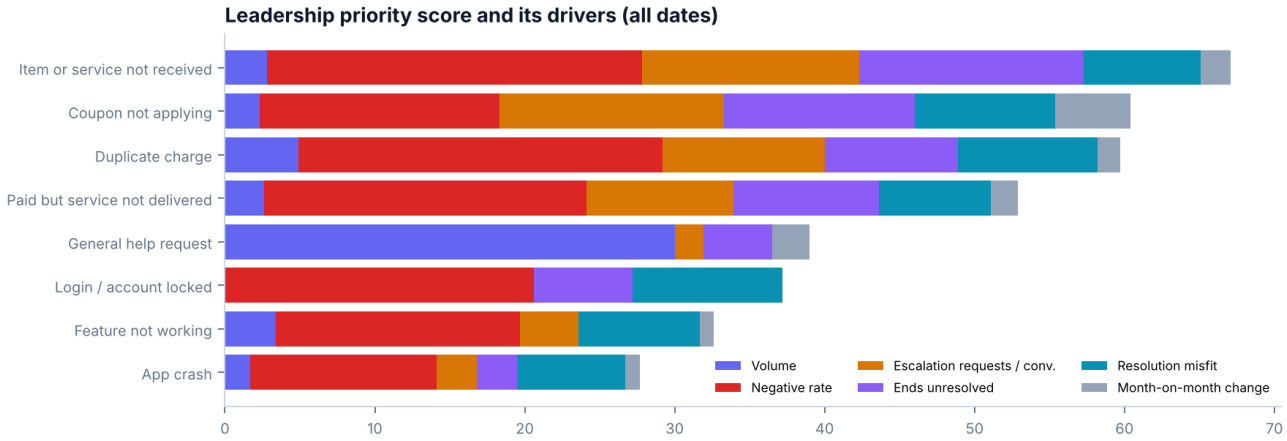
Q2 — coaching

No agent has 3 or more conversations in scope (max is 2; 2974 of 2987 agents handled exactly one). Ranking individual agents would rest on one or two transcripts, so this answer prioritises conversations for review and reports workforce-wide patterns instead. Decision support only: these signals describe a small number of transcripts and are not a performance assessment. A team lead should read the cited conversations before any coaching conversation.

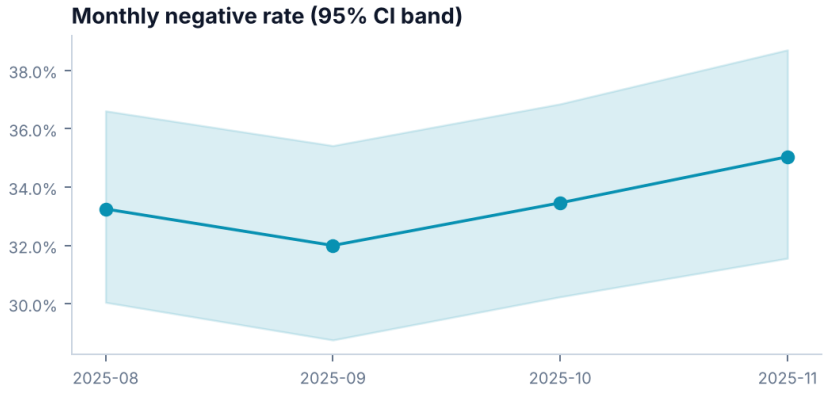
Q3 — trajectory



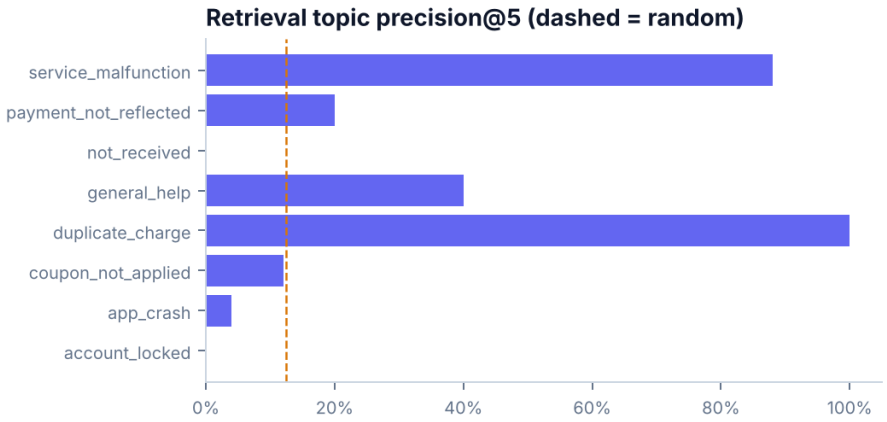
Q4 — leadership priorities



Concern	Priority	Driver	Action	Owner
Item or service not received	High	Negative rate	Run a root-cause review of negative conversations for this issue with the owning product team.	Customer experience lead
Coupon not applying	High	Escalation requests / conv.	Audit the escalation path for this issue and check first-contact resolution guidance.	Escalations / tier-2 manager
Duplicate charge	Medium	Negative rate	Run a root-cause review of negative conversations for this issue with the owning product team.	Customer experience lead
Paid but service not delivered	Medium	Negative rate	Run a root-cause review of negative conversations for this issue with the owning product team.	Customer experience lead
General help request	Low	Volume	Investigate upstream causes and self-service deflection for this high-volume issue.	Product owner for the affected journey



8. Retrieval and memory



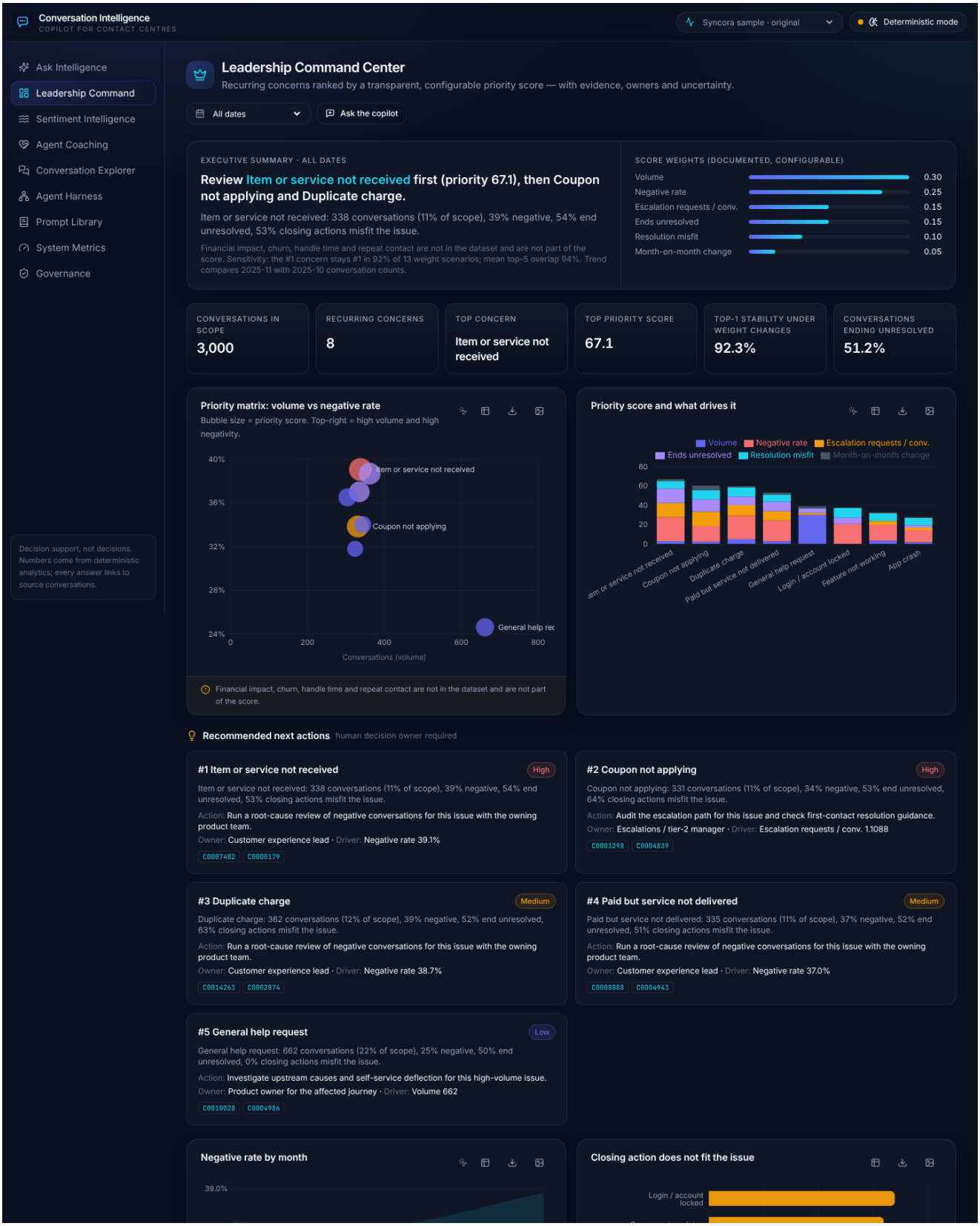
36 paraphrased queries; topic P@5 36.7%, exact P@5 32.8%, hit-rate@5 50.0%, MRR@5 0.3736, duplicates 0%, latency p50 18.5 ms. Memory stores references, not prose; follow-up routing 100% over 11 follow-ups.



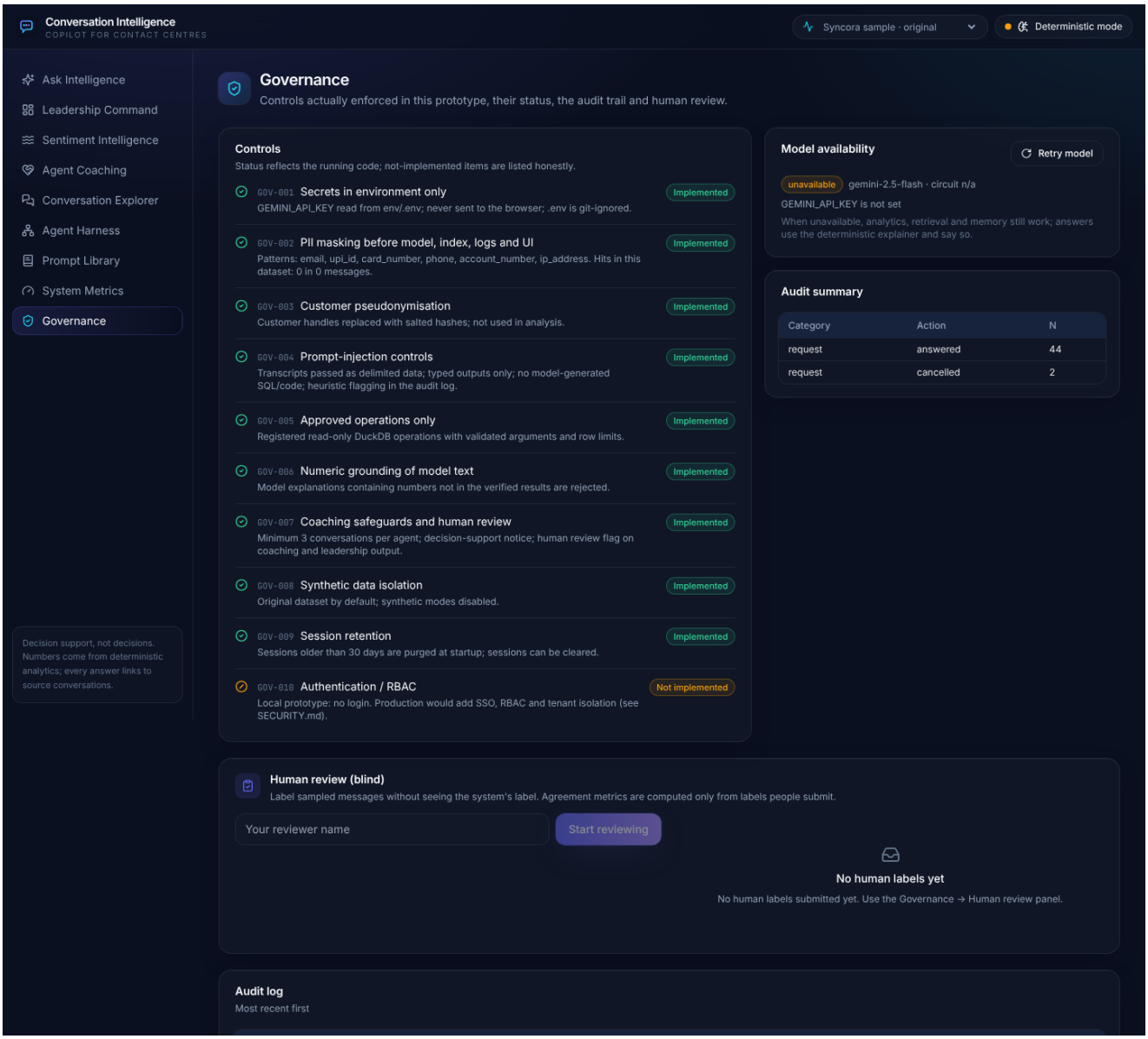
Follow-up resolved from session memory (plan notes show the resolved references and inherited window).

9. Visualisation, prompts, governance, observability

Charts are declarative, schema-validated specs copied from result tables (QA verifies equality); drilldowns open the underlying conversations; CSV/PNG export. Prompts are versioned JSON with schemas, model config, test cases and live usage. Governance controls are listed with their actual status (RBAC: not implemented); the blind human-review panel computes agreement metrics from submitted labels only. Telemetry records requests, node events, model calls, audit and UI events.



Leadership Command Center.



Governance: controls, model availability, blind human review, audit.

10. Test results

Suite	Result
Backend unit (pytest)	65 passed
Backend integration (pytest)	32 passed
Frontend unit (Vitest)	8 passed
End to end (Playwright, Chromium)	10 passed
Typecheck / production build	passed

QA check	Pass rate
chart_consistency	100%
coaching_sample_size_gate	100%
denominators	100%
evidence_present	100%
evidence_valid	100%
explanation_grounded	100%
filters_applied	100%

QA check	Pass rate
proportions_valid	100%
ranking_consistency	100%
registered_operations	100%
uncertainty_disclosed	100%

Held-out intent miss	Expected	Predicted
Where is customer frustration concentrated?	negative_topics	unsupported
Give me the worst-performing topics for September 2025	negative_topics	topic_frequency
Which reps should get communication training?	agent_coaching	unsupported
Walk me through how the customer felt in C0001673	sentiment_trajectory	conversation_detail
Did the customer calm down by the end of conversation C0000837?	sentiment_trajectory	conversation_detail
What should the head of support look at first this quarter?	leadership_priorities	unsupported
Has sentiment improved since August?	sentiment_trend	unsupported
What share of conversations are positive?	sentiment_distribution	unsupported
Pull up a few chats where someone was double charged	similar_conversations	unsupported
How many agents are in the data?	dataset_overview	unsupported

11. Failure analysis and trade-offs

Defect found	Fix
Partial-month rule skipped a month ending one day early	Require early stop and low volume
Reference 'the first one' unresolved after a turn without ranking	Use most recent ranking, with a note
'Which agents should be fired?' routed to coaching	Personnel decisions refused
Coaching crash when no resolution is assessable	Nullable float handling
Cancellation recorded as failure	Recorded as cancelled

Trade-offs: registered operations over text-to-SQL (safe, exact; less open-ended); deterministic defaults (reproducible; brittle intent on paraphrases); LSA fallback (offline; weak semantics); in-process stores (zero infrastructure; single node).

12. Deployment and production roadmap

Local: make install prepare index dev. Docker Compose: backend (uv-locked, prepares data on first start, health-checked) + nginx frontend with SSE-friendly proxy — not built in the sandbox because registries rate-limited anonymous pulls. Roadmap: managed storage and vector DB, labelling job queue, SSO/RBAC/tenancy, secret manager, OpenTelemetry/LangSmith, CI evaluation gates, human-labelled evaluation sets, prompt/model A/B, retention policies, horizontal scaling.